DAY 31 | POST-READ

# LLMOps and Escalations: Recap and Beyond

What we covered today, plus ideas to keep developing.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

## Part A: Summary of today

| Topic | What to remember |
|---|---|
| LLM lifecycle | Scope, build, evaluate, deploy, operate, and loop back. |
| Why LLMOps differs | Variable outputs, provider model updates, per-token cost, silent drift. |
| Failure modes | Rate limits, timeouts, model update regressions, cost spikes, retrieval breaks, attacks. |
| Backoff with jitter | Wait 2^attempt plus a random amount; retry only transient errors; cap attempts. |
| Fallbacks | Backup model, cached answer or graceful message. |
| Circuit breaker | After N failures, stop calling for a cool-down period, then test again. |
| Severity | SEV1 to SEV4 by impact; use the customer's definitions where they exist. |
| Escalation | What, impact, evidence, actions so far, ask, next update time. |
| Emotional intelligence | Self-awareness, self-regulation, motivation, empathy, social skill. |

## Part B: Going beyond today

### 1. Blameless post-incident reviews

After an incident, write a timeline, the root causes, what went well, and actions with owners. Focus on systems, not people. Teams that do this learn faster and hide fewer problems.

### 2. Pin model versions

Where the provider offers dated model versions, use them in production and upgrade deliberately after running your golden set, instead of letting an alias change underneath you.

### 3. Idempotent retries

Retrying a request that triggers an action (send email, issue refund) can duplicate it. Use idempotency keys or check whether the action already happened before retrying.

### 4. Status pages and runbooks

A simple status page and a runbook for each known failure ("429 storm", "provider outage") turn panic into a checklist.

### 5. Managing your own stress

Before a tense call: breathe slowly for a minute, write the three points you must make, and decide your next update time. Afterwards, take a short break before the next task.

## Part C: Reflect

- Which failure mode would hurt your capstone project most, and how would you detect it?
- Rewrite a defensive sentence you have used at work into an empathetic one.
- What is your personal early warning sign of stress?
- Draft the escalation message for the incident in today's simulation.

## Part D: Practice ideas for tonight

- Add a circuit breaker to your Prompt Playground or Task Board LLM calls.
- Write a runbook for a 429 rate-limit storm.
- Practise the escalation structure on a real issue from your work, without sending it.

**Coming up on Day 32:** MLOps vs LLMOps, fine-tuning, versioning, monitoring, and cost and latency. There is also a 1.5 hour assessment.

*Otsukaresama deshita (お疲れ様でした): thank you for your hard work today*